Thinh Nguyen Nhu

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Introduction

As a science-oriented person, my long-term vision is to develop intelligent systems and structures that learn from the fundamental principles of physics, biology, and engineering. My research interests focus on applied machine learning, particularly Physics-Informed Machine Learning (PIML) and Generative AI for scientific discovery and inverse problems. My ultimate goal is to build physics-grounded AI models that enable the understanding, design, and control of next-generation smart structures.

Education

Hanoi University of Science and Technology (HUST) Hanoi, Vietnam
B.Sc. in Mechatronics Engineering *10/2022 – 8/2026*
GPA: 3.69 / 4.0
Thesis: Modeling and Inverse Design of Kresling Origami via Physics-Informed Neural Networks

Research & Projects

AiRA Lab (AiRA), HUST Hanoi, Vietnam
Research Assistant *09/2026 – Present*
- Investigate Physics-Informed Machine Learning (PIML) methods for scientific and engineering applications.
- Develop PINN architectures for multi-solution inverse design of Kresling origami structures.
- Implement PINN-based frameworks for structural analysis and design optimization of composite laminates.

Origami Technology Lab, HUST Hanoi, Vietnam
Undergraduate Student *07/2025 – 08/2026*
- Modeled nonlinear dynamic behaviors of Kresling origami and constructed bifurcation diagrams under harmonic excitation.
- Formulated Physics-Informed Neural Network (PINN) algorithms for the inverse design of Kresling structures as part of undergraduate thesis research.

Project XYaZ Lab, HUST Hanoi, Vietnam
Undergraduate Student *10/2024 – 06/2025*
- Modeled and simulated the dynamics of a 3-RRR planar parallel robot using MATLAB/Simulink.
- Evaluated trajectory-tracking algorithms for parallel mechanisms.

Publications

Gradient-Based Inverse Design of Kresling Origami via Neural Surrogate Modeling, Thanh Trung Vuong, Duc Thu Nguyen, Nguyen Nhu Thinh, Thao Thai Phuong*, Chien Truong Quoc*; *Robotics and Mechatronics, ISRM 2026*; 03 Jul 2026, [10.1007/978-3-032-29469-2_39].

A Comprehensive Analysis of Global Dynamics and Bifurcation of a Kresling Origami Structure, Minh-Tuan Nguyen-Thai; Thinh Nguyen Nhu; Ngoc-Vu Le-Nguyen; Chien Truong-Quoc*; Thao Phuong Thai*; *submitted*; 2026, [10.2139/ssrn.6721329].

Dynamic Behavior of Kresling Origami under Harmonic Excitation: A Five-Parameter Energy

Framework, Thinh Nguyen Nhu, Chien Truong Quoc*, Thao Thai Phuong*; *The 8th International Conference on Engineering Mechanics and Automation (ICEMA25)*; 14 Nov 2025, [10.15625/vap.2026.0031].

Skills

Programming: Python, PyTorch, JAX, MATLAB/Simulink, C/C++

Machine Learning: Physics-Informed Machine Learning (PIML), Operator Learning, Kolmogorov–Arnold Networks (KAN), Generative Modeling (VAE, Diffusion), CNN

Mathematical Tools: Optimization, Mathematical modeling of dynamical systems, Numerical simulation, Nonlinear system analysis, FEA

Misc.: Academic research, Reading scientific literature and writing in English, LaTeX typesetting, research presentations

Achievements

Academic Encouragement Scholarship for semesters 2023.2, 2024.1, 2024.2, and 2025.2 – HUST.

Outstanding Creative Idea Award - 43rd HUST Annual Student Research Conference (SVNCKH), HUST. (2026)

VPMS Award - Graduation Thesis with Scientific Value, HUST. (2026)

Certificate of Merit for Outstanding Student, Academic Year 2024–2025, HUST. (2025)

Competitions

Biohub - Cell Tracking During Development (Kaggle Competition) 9/2026
Developed a deep learning pipeline for cell segmentation and multi-object tracking.

References

Thao Thai Phuong, PhD Lecturer
thao.thaiphuong@hust.edu.vn *School of Mechanical Engineering, HUST*

Chien Truong-Quoc, PhD Lecturer
chien.truongquoc@hust.edu.vn *School of Mechanical Engineering, HUST*